

EDINBURGH COURSEWORK / REVISED EXPOSITION

Accuracy and Stability in Numerical ODEs

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Three revised proof-based studies

Based on individually authored Edinburgh assignments, February–March 2023. Revised 25 September 2026, with AI-assisted mathematical and editorial checking. This edition corrects and expands the historical coursework; the results are standard mathematical exposition.

1 Accuracy, stability, and order conditions

1.1 Euler's method: finite-time convergence and long-time instability

Source and revision. This exposition revises Problem 3 of the archived NODE Assignment 1 (February 2023). The product formula and the distinction between exceptional and ordinary step sizes are present in the original coursework. The explicit divergence bound and the finite-horizon error estimate below were added in the 2026 revision. These are standard numerical-analysis exercises, not new research results.

Consider the scalar initial-value problem

$$y'(t) = -ty(t), \quad y(0) = y_0 > 0, \quad t \geq 0.$$

Direct differentiation verifies that its solution is $y(t) = y_0 \exp(-t^2/2)$, which tends to zero as $t \rightarrow \infty$. On the uniform grid $t_n = nh$, with $h > 0$, forward Euler gives

$$u_0 = y_0, \quad u_{n+1} = (1 - nh^2)u_n.$$

The symbols u_n and $y(t_n)$ will distinguish the numerical and exact solutions throughout.

Proposition 1.1 (Long-time dichotomy at fixed step size). *For every integer $n \geq 1$,*

$$u_n = y_0 \prod_{k=0}^{n-1} (1 - kh^2).$$

If $h^{-2} = m$ is a positive integer, then $u_n = 0$ exactly when $n \geq m + 1$. If h^{-2} is not a positive integer, then $|u_n| \rightarrow \infty$ as $n \rightarrow \infty$.

Proof. The product formula follows by induction from the Euler recurrence; the $k = 0$ factor is one. A factor vanishes if and only if $k = h^{-2}$. Consequently, if $h^{-2} = m \in \mathbb{N}$, the first zero occurs in u_{m+1} , and every subsequent iterate is zero.

In the other case no factor vanishes. Choose an integer $N \geq 3/h^2$. For every $k \geq N$,

$$|1 - kh^2| = kh^2 - 1 \geq 2.$$

Since $u_N \neq 0$, for $n \geq N$ we obtain

$$|u_n| = |u_N| \prod_{k=N}^{n-1} |1 - kh^2| \geq |u_N| 2^{n-N} \rightarrow \infty.$$

This tail bound, rather than the growth of a single factor alone, establishes divergence of the whole product. $\square$

Long-time instability at a fixed h does not contradict convergence on a fixed time interval as $h \rightarrow 0$. The following estimate makes the distinction explicit.

Proposition 1.2 (A finite-horizon error bound). *Fix $T > 0$ and let $M_T = \max_{0 \leq t \leq T} |y''(t)|$. If $hT \leq 1$, then, at every grid point with $nh \leq T$,*

$$|u_n - y(t_n)| \leq \frac{TM_T}{2} h.$$

In particular, forward Euler converges uniformly on the grid points in $[0, T]$ with first-order accuracy.

Proof. Taylor's theorem gives, whenever $t_{n+1} \leq T$,

$$y(t_{n+1}) = (1 - ht_n)y(t_n) + \delta_n, \quad |\delta_n| \leq \frac{1}{2}M_T h^2.$$

For $e_n = u_n - y(t_n)$, subtraction yields $e_{n+1} = (1 - ht_n)e_n - \delta_n$. The assumption $hT \leq 1$ implies $0 \leq 1 - ht_n \leq 1$; hence

$$|e_{n+1}| \leq |e_n| + \frac{1}{2}M_T h^2.$$

Starting from $e_0 = 0$ and iterating gives $|e_n| \leq nM_T h^2/2 \leq TM_T h/2$. The constant is finite because $y''(t) = (t^2 - 1)y_0 e^{-t^2/2}$ is continuous on $[0, T]$. $\square$

Interpretation. The two statements take different limits. In the first, h is fixed and t_n increases without bound. In the second, T is fixed and h decreases. Here the coefficient in the ODE grows with time, so a single fixed step size does not ensure decaying iterates over arbitrarily long intervals.

1.2 Order and stability of two-stage explicit Runge–Kutta methods

Source and revision. This section revises NODE Assignment 3, Problem 1 (March 2023), with the explicit midpoint example from Assignment 2, Problem 2. The 2026 revision states the order assumptions, supplies the local Taylor argument, and distinguishes the real-axis stability interval from the full complex stability region.

For a sufficiently smooth scalar ODE $y' = f(t, y)$, consider the family

$$\begin{aligned} k_1 &= f(t_n, u_n), \\ k_2 &= f(t_n + ch, u_n + ch k_1), \\ u_{n+1} &= u_n + h(b_1 k_1 + b_2 k_2), \end{aligned}$$

where $c, b_1, b_2 \in \mathbb{R}$ are fixed, with $c \neq 0$. The equality between the time shift and the internal stage coefficient is part of the specified method.

Proposition 1.3 (Second-order conditions and amplification factor). *Assume*

$$b_1 + b_2 = 1, \quad b_2 c = \frac{1}{2}.$$

The method has classical consistency order exactly two. For the test equation $y' = \lambda y$, $\lambda \in \mathbb{C}$, its amplification factor is

$$R(z) = 1 + z + \frac{1}{2}z^2, \quad z = h\lambda.$$

For example, $c = 1/2$, $b_1 = 0$, $b_2 = 1$ gives the explicit midpoint method.

Proof. Take one step starting from the exact value $y = y(t)$. If f is C^2 on a neighbourhood of the point and stages under consideration, Taylor expansion gives

$$f(t + ch, y + chf(t, y)) = f + ch(f_t + f_y f) + O(h^2),$$

where the functions and derivatives on the right are evaluated at (t, y) . The numerical one-step value is therefore

$$y + h(b_1 + b_2)f + h^2 b_2 c(f_t + f_y f) + O(h^3).$$

On the other hand, $y'' = f_t + f_y f$, so the exact value is

$$y(t + h) = y + hf + \frac{1}{2}h^2(f_t + f_y f) + O(h^3).$$

The stated coefficient conditions make the one-step defect $O(h^3)$.

For the linear test equation, $k_1 = \lambda u_n$ and $k_2 = \lambda(1 + cz)u_n$, giving

$$u_{n+1} = [1 + (b_1 + b_2)z + b_2 cz^2]u_n = R(z)u_n.$$

Since $e^z - R(z) = z^3/6 + O(z^4)$, the method fails the third-order condition already on $y' = y$. It thus has order exactly two, not merely at least two. $\square$

Proposition 1.4 (Real and imaginary axes). *Let $\mathcal{S} = \{z \in \mathbb{C} : |R(z)| \leq 1\}$ be the closed absolute-stability region. Then*

$$\mathcal{S} \cap \mathbb{R} = [-2, 0], \quad \mathcal{S} \cap i\mathbb{R} = \{0\}.$$

Strict decay on the real axis, $|R(x)| < 1$, holds exactly for $-2 < x < 0$.

Proof. For real x ,

$$R(x) = \frac{1}{2}[(x+1)^2 + 1] > 0.$$

Consequently $|R(x)| \leq 1$ is equivalent to $(x+1)^2 \leq 1$, which is equivalent to $-2 \leq x \leq 0$. Strict inequalities give $(-2, 0)$. For $\omega \in \mathbb{R}$,

$$|R(i\omega)|^2 = \left(1 - \frac{1}{2}\omega^2\right)^2 + \omega^2 = 1 + \frac{1}{4}\omega^4.$$

This is at most one only when $\omega = 0$. $\square$

Interpretation. For real $\lambda < 0$, decaying iterates require $0 < h < -2/\lambda$. At the nonzero boundary point $h\lambda = -2$, $R(-2) = 1$, so iterates remain bounded but do not decay. For a purely imaginary nonzero λ , the exact solution has constant magnitude, whereas these methods increase its magnitude at each step. These conclusions concern the specified *two-stage* family; they do not describe every higher-stage method of order two.

1.3 A multistep scheme: zero-stability does not imply second order

Source and revision. This section reconstructs the method from the coefficients explicitly listed in NODE Assignment 3, Problem 3, pages 3–4 (March 2023). The original second-order conclusion is corrected in this 2026 revision. The analysis applies to those recorded coefficients; the original instructor's question is not available here to resolve any possible transcription error in the scheme itself.

Consider

$$u_{n+2} - \frac{4}{3}u_{n+1} + \frac{1}{3}u_n = h\gamma f(t_{n+1}, u_{n+1}), \quad \gamma \in \mathbb{R}. \quad (1)$$

For a smooth function y , define its *unnormalized residual* by

$$D_h[y](t) = y(t + 2h) - \frac{4}{3}y(t + h) + \frac{1}{3}y(t) - h\gamma y'(t + h).$$

Order p means $D_h[y](t) = O(h^{p+1})$ for all sufficiently smooth y . Dividing this residual by h gives the alternative normalized local truncation error convention.

Proposition 1.5 (A stable recurrence with incompatible order conditions). *The scheme satisfies the root condition for zero-stability for every γ . It is consistent if and only if $\gamma = 2/3$, in which case its order is exactly one. No value of γ gives order two.*

Proof. The first characteristic polynomial is

$$\rho(\xi) = \xi^2 - \frac{4}{3}\xi + \frac{1}{3} = (\xi - 1)(\xi - \frac{1}{3}).$$

Its two roots are simple, both have modulus at most one, and only $\xi = 1$ lies on the unit circle. This is precisely the root condition. Equivalently, a solution of the homogeneous perturbation recurrence $v_{n+2} - \frac{4}{3}v_{n+1} + \frac{1}{3}v_n = 0$ has the form $v_n = A + B3^{-n}$, so the influence of its starting values remains bounded as the number of steps grows. This part is independent of γ .

For the order calculation, Taylor expansion at t yields

$$D_h[y](t) = h\left(\frac{2}{3} - \gamma\right)y'(t) + h^2\left(\frac{4}{3} - \gamma\right)y''(t) + O(h^3).$$

The coefficient of $y(t)$ vanishes automatically. First-order consistency requires the coefficient of $hy'(t)$ to vanish, so $\gamma = 2/3$. With this value,

$$D_h[y](t) = \frac{2}{3}h^2y''(t) + O(h^3).$$

This is $O(h^2)$ for smooth y , but it is not generally $O(h^3)$: for $y(t) = t^2$ the residual is exactly $4h^2/3$. Thus the order is exactly one.

Order two would require *both* $\gamma = 2/3$ and $\gamma = 4/3$, which is impossible. In particular, choosing $\gamma = 4/3$ to cancel the second-order coefficient does not preserve first-order consistency: for $y(t) = t$,

$$D_h[y](t) = -\frac{2}{3}h \neq O(h^2).$$

This simple test function exposes the incorrect second-order conclusion. □

Interpretation. Order conditions must hold simultaneously at every lower degree. Satisfying the condition involving a quadratic test function cannot replace the condition involving a linear one. The root condition addresses the propagation of starting perturbations; consistency addresses whether the scheme approximates the intended differential equation. Neither property should be substituted for the other.